ABDELWAKIL KHAMSI

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PROFILE

Mechanical Engineering student at Bournemouth University building a torque-controlled, dynamically-walking bipedal robot for my MSc dissertation — spanning mechanical design, embedded electronics, and robotics software (ROS2, control theory, reinforcement learning). Previously contributed to the design and validation of a hydrogen fuel cell system for a heavy-duty truck at Segula Technologies, working with Stellantis across a cross-functional team in the US, Germany, France, and Romania. I've since broadened that foundation with maintenance engineering internships across food processing and fertiliser manufacturing.

I'm drawn to the intersection of mechanics, electronics, and control theory — where a CAD model only becomes a machine once the right controller and code bring it to life. That cross-domain fluency, paired with a genuine enjoyment of building things that actually move, is what I bring to every project.

TECHNICAL SKILLS

- **SOFTWARE:** Catia v5/v6, SolidWorks, Fusion 360, Autodesk Inventor, ANSYS Mechanical, Abaqus
- **PROGRAMMING:** Python, C++.
- **CMMS:** Optimaint, Oracle.
- **ROBOTICS & CONTROL:** ROS2, Gazebo, Pinocchio, Isaac Lab, CAN bus, ODrive motor control, Embedded systems, Linux, Trajectory Planning, Inverse Kinematics.
- **ENGINEERING SKILLS:** Rapid Prototyping, 3D Printing, Tolerance stack Analysis, Mechanical drawings, Geometric dimensioning and tolerancing, Instrumentation Drawings, FMEA, Material Behaviour, Life cycle analysis, DFM.

Master of Science in Mechanical Engineering — **September 2025 - September 2026**

Bournemouth University, England

- Relevant coursework in Advanced Structural Mechanics, Failure Analysis and Prevention, Robotic Control Design and Model Based Engineering.

Master of Science in Mechanical Engineering: Maintenance and Quality — **September 2021 - July 2023**

Polytechnic School of Agadir, Morocco

- Thesis on "Improving the Availability of the PS III Sulphuric Acid Line through the Implementation of a Shell Replacement Procedure."

Bachelor of Science in Maritime Mechanical Engineering — **September 2018 - July 2021**

Higher Institute of Maritime Fisheries, Morocco

- Thesis on "Design and Implementation of a PID Temperature Control System for a Sardine Canning Cooking Installation."

WORK EXPERIENCE

Design Release Engineer, Segula Technologies — **November 2023 - August 2024**

- Contributed to the development of a hydrogen fuel cell system for a heavy-duty truck for Stellantis, collaborating with a cross functional team across the US, Germany, France, and Romania.
- Simplified system architecture by supporting the removal of the turbine bypass, reducing cost by €32.50 per part and lowering system complexity without compromising durability.
- Supported supplier issue resolution on components, such as fuse integration, temperature sensors, engine mounts, and subsystem mounting points.
- Enhanced vehicle safety by validating the "eBurn" function, an energy-dissipation strategy that prevents brake overheating and maintains regenerative braking on long downhill grades.
- Participated in testing and validation activities, analysing failures (e.g. Hydrogen inlet pressure valve) and helping define improvement measures.

Graduate Mechanical Engineering Internship, OCP Group **February 2023 - May 2023**

- Led a root cause analysis study of critical production line failures, applying fishbone diagrams and 5 why's to identify and eliminate recurring issues.
- Created a 3D model of the Heat Recovery System Tower in SolidWorks and ran a static structural simulation in ANSYS Mechanical to evaluate external constraints.
- Developed and implemented a streamlined shell replacement procedure for the HRS Tower, reducing downtime and improving operational efficiency.
- Delivered a 15% increase in overall availability of the PS III sulphuric acid line by integrating simulation insights with optimised maintenance strategies.

Design & Manufacturing Engineering Internship, ENSA Mechanical Laboratory **June 2022 - August 2022**

- Designed a 3D model of an educational torsion test bench in SolidWorks, applying reverse engineering principles such as systematic decomposition and functional analysis to replicate and enhance the original design.
- Upgraded the measurement system by replacing the external comparator with a graduated pulley, improving accuracy and ease of use.
- Operated both conventional and CNC machining equipment to manufacture and assemble project components.
- Achieved an 80% reduction in setup time for the test bench through the redesigned measuring system.

PROJECTS

Bipedal Robot — MSc Dissertation Project (RABBIT–class Planar Biped)
February 2026 – Present

Funded by [Grant Name]; manuscript in preparation

Designing and building a 5-link, four-actuator planar bipedal robot for my MSc dissertation, extending a published single-leg design with custom mechanical, electrical, and software architecture.

- Built full ROS2-based robot architecture from scratch, including URDF kinematic modeling, trajectory planning, and inverse kinematics for the leg.
- Integrated brushless motor actuation (ODrive controllers, CAN bus) with closed-loop position control, validated through iterative prototyping, 3D printing, and bench testing.
- Progressing toward dynamic walking via torque-based control, comparing classical control (Hybrid Zero Dynamics) against reinforcement learning (Isaac Lab) under shared performance metrics.

Life Cycle Management Project **September 2025 – January 2026**

Improving the carbon footprint of a Roll-A-Car 740 Mechanical Jack

Life Cycle Management project focused on improving the environmental impact of a mechanical lifting jack through quantified life cycle analysis and redesign on SolidWorks.

- Established a baseline carbon footprint of around 560 kg CO_2, primarily driven by material choice and manufacturing processes.
- Ran a topology optimisation on ANSYS while preserving 55% to 60% of the starting mass.
- Re ran a static structural analysis to validate that the performance was maintained post redesign.
- Reduced total carbon footprint to 440 kg CO_2, achieving an approximate 22% reduction.

3D-Printed Robotic Arm (3DOF → 5DOF → 6DOF) **October 2024 – August 2025**

- Designed and iteratively developed a robotic arm from 3DOF to a 6DOF system, focusing on precision, torque, and efficiency through evolving drive technologies (planetary gearboxes → hybrid cycloidal–belt transmissions).
- Optimised manufacturability and assembly, reducing hardware requirements by using standard ball bearings (608-2z), stepper motors (Nema 17), and bolts. Ensured parts were low cost, easy to print, and reliable on consumer-grade 3D printers.
- Implemented a two-layer control system: a high-level layer (ROS + MoveIt in a VM) generated joint angles from user-defined RViz poses via inverse kinematics, while a low-level Python layer translated those angles into real-time stepper motor commands.

ADDITIONAL INFORMATION

- **Languages**: English (Professional working proficiency), French (fluent), Arabic (native).
- **International experience:** Academic and professional experience across Morocco, the UK, and multinational engineering teams in Europe and North America.
- **Driving licence:** International Driving Permit valid until September 2026 and currently working on getting a full UK licence.